Centro de Investigación en Complejidad Social · Universidad del Desarrollo · Chile

Networks Decide for Us

Imputed homophilic networks and the structural premium of diffusion

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Collective behavior spreads on networks

- Collective behaviors — innovations, mobilizations — spread as **contagions** on social networks.
- Real human networks have a specific texture: we cluster with people like us — **multidimensional homophily** [1].
- So a natural question: how much of *where* a behavior ends up is about *who people are*, and how much is about *how they are wired together*?

The gap

- Diffusion ABMs run on *made-up* networks (small-world, scale-free) [2].
- Rich surveys (ATP, GSS) measure *propensities* but carry *no* relational data.
- Threshold models often assume *all* adoption needs peer influence — and when weak-interest groups still tip, the usual explanation is *stochastic noise* that spontaneously instigates a cascade [8].

We ask whether **measured preference and real structure** — not noise, not synthetic graphs — can do that work.

Where we go

Two moves

- **Impute** plausible whole-networks onto real survey respondents from empirical homophily — going from McPherson & Smith's latent "context" term [4] to **full simulable networks**.

- **Measure** the causal premium of that structure on diffusion — and ask *what about the structure* actually matters.

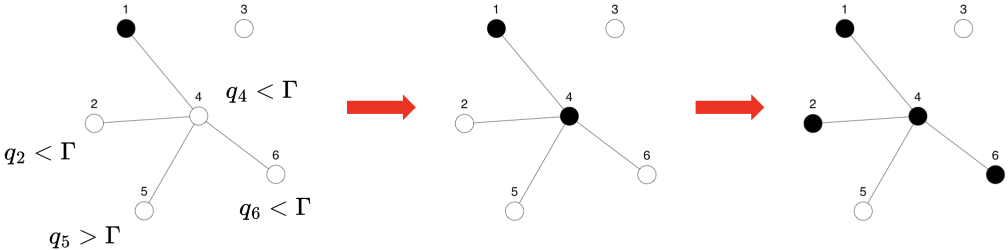
Holding individual propensities *fixed*, does the **shape of the network alone** — not who its members are — decide *how much* collective behavior spreads?

The model

Adoption, part 1 — Rational Choice

- A behavior has an *Intrinsic Utility Level* **IUL**, $\Gamma \in [0, 1]$.
- Each person has a *Minimum Utility Requirement* **MUR**, $q_i \in [0, 1]$, representing their aversion to adopting the behavior.

An agent adopts on its own iff $\boxed{q_i \leq \Gamma \implies a_i = 1}$



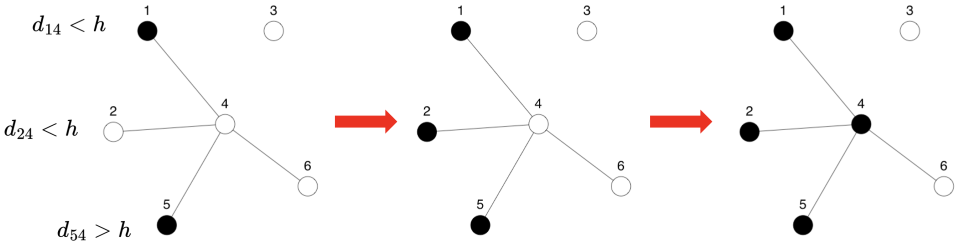
This channel is **topology-blind** — it does not need the network at all.

Adoption, part 2 — Selective Social Influence

Otherwise, an agent adopts by peer contagion if $\tau_i \leq \tilde{E}_i$ [3] — but only peers who are socially *close* [6] count:

$$\tilde{E}_i \equiv \frac{\sum_{j\neq i} \mathbf{X}_{ij}\, \tilde{a}_j}{\sum_{j\neq i} \mathbf{X}_{ij}}, \qquad \tilde{a}_j = 1 \iff a_j = 1 \wedge U_{ij} \leq \sigma(MSP - d_{ij})$$

- d_{ij}: social (Blau-space) distance; **MSP** (Maximum Social Proximity) sets how far across demographic lines influence can reach.



Imputing structure & running the model

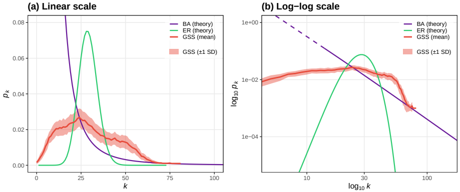
Imputing a network onto a survey

Homophily strengths $\hat{\boldsymbol{\beta}}$ [6,4] → Construct realistic networks → Add propensity scores q_i to each node

Tie probability is a function of social distance (age, sex, education, race, religion):

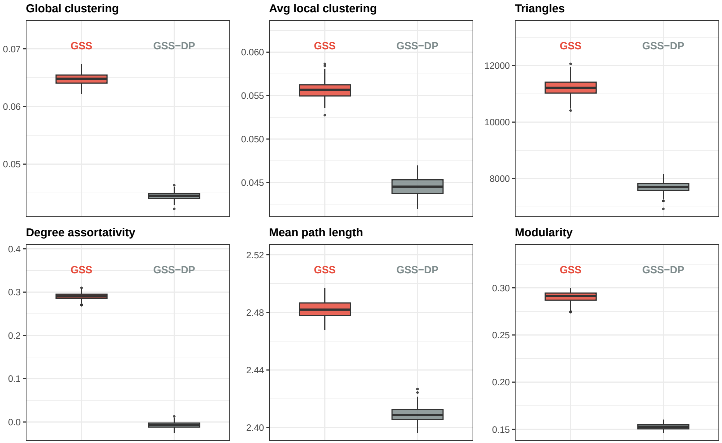
$$\text{logit } \Pr(Y_{ij} = 1) = \alpha + \boldsymbol{\beta}^\top \mathbf{d}_{ij}$$

$\hat{\boldsymbol{\beta}}$ comes from GSS confidant data [6]; propensities q_i come from real survey answers (innovation, collective action).



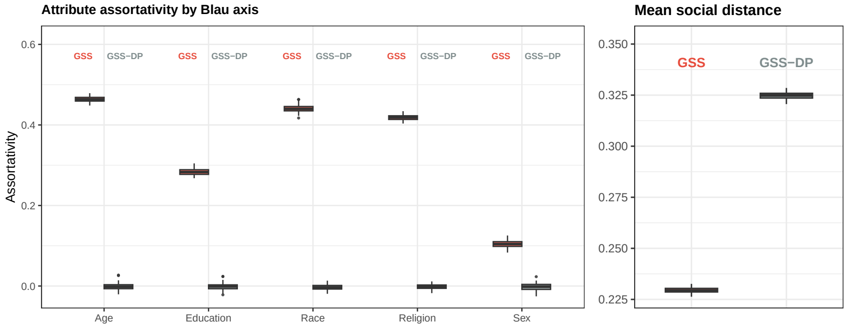
Right-skewed degree structure that Erdös-Rényi / Barabási-Albert models fail to match jointly.

What does “breaking structure” destroy? — topology



100 plausible (GSS) vs. 100 degree-preserving randomized (GSS-DP). Clustering -31%, triangles -31%, modularity -47% — yet path length barely moves, and degree/density/giant-component are identical. The topology degrades, but not uniformly.

What does “breaking structure” destroy? — homophily



Homophily is annihilated: every Blau-axis assortativity collapses to ≈ 0, and the model’s own tie distance (d_{ij}) rises **+42%** — neighbors go from *similar* to *random*. This is the prime suspect behind the premium. All $p < 10^{-30}$.

Comprehensive simulation design

Hybrid social contagion via `netdiffuseR` [7], over plausible **GSS** vs. degree-preserving randomized **GSS-DP** topologies.

- **IUL** (Γ): 41 levels $[0, 1]$. **MSP** (h): 13 levels $[0, 1]$.
- **Thresholds** $\tau_i \sim \mathcal{N}(\mu, \sigma)$: $\mu \in \{.3, .4, .5, .6\}$, $\sigma \in \{.08, .12, .16, .20\}$.
- **Seeding**: random, central, marginal, closeness, eigenvector.

Total: $41 \times 13 \times 4 \times 4 \times 5 \times$ (runs) $\approx$ 4×10^6 **simulations**

We then fit *Big Additive Models* (BAM) to isolate the structural effect:

$$\operatorname{logit}\left[\mathbb{E}(\Phi)\right] = \beta_0 + \mathbf{Z}\boldsymbol{\beta} + \beta_{\text{DP}}\,\mathbb{I}_{\text{DP}} + s(\text{IUL}, \text{MSP})$$

Results

Result 1 — A structural premium

A Big Additive Model isolates the marginal effect of destroying empirical structure:

$$\boxed{\beta_{\text{DP}} = -0.850^{***} \quad (\text{OR} = 0.43)}$$

$\rightarrow$ randomization cuts adoption odds by $\sim$**57%**

Same people, same propensities — only the **wiring** changed. Homophilic clustering lets social reinforcement build inside local echo chambers.

Total adoption (GSS)	Estimate	OR
(Intercept)	6.534^{***}	—
τ_{mean}	-6.944^{***}	—
τ_{sd}	5.942^{***}	—
Seeding (ref: Random)		
Central	0.120^{***}	1.127
Closeness	0.119^{***}	1.126
Eigenvector	0.118^{***}	1.125
Marginal	-0.067^{***}	0.935
Topology (ref: Plausible)		
Deg.-pres. random	$\mathbf{-0.850}^{***}$	**0.427**
Deviance explained: 82.2%	$N = 4{,}093{,}440$	

$^{***}p < 0.001$. OR $= \exp(\beta)$.

Conclusions

Networks decide for us.

1. **A method:** impute **simulable whole-networks** onto survey respondents from empirical homophily — reusable wherever you have propensities but no ties.
2. **A finding:** holding propensities fixed, empirical structure yields a $+57\%$ **adoption premium**.
3. **A mechanism:** what "breaking structure" destroys is, above all, **homophily** — neighbors stop being similar.

Unlike Macy & Evtushenko [8], the idiosyncrasy that drives this is **measured preference**, not stochastic noise: *the same agents, on a different topology, produce a different society.*

References

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Thanks!

Scan: preprint draft (Zenodo)

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`https://doi.org/10.5281/zenodo.19360209`

Backup — the *shape* of change also differs

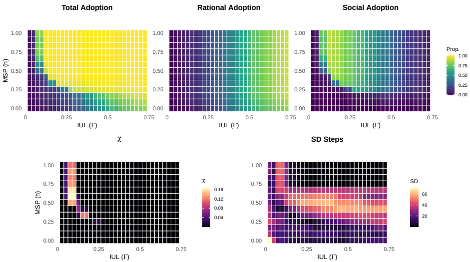
Beyond *how much*, structure governs *how* adoption tips:

- Plausible (GSS) → **continuous, predictable** transition.
- Randomized (GSS-DP) → **abrupt** regime shift.

Near the tipping point the GSS networks show **mean-field universality**; the randomized ones do not [9].

Critical exponents (mean over IUL):

	Plausible (GSS)	Random (GSS-DP)
Susceptibility γ	≈ 0.91	≈ 3.9
Critical isotherm δ	≈ 2.85	≈ 2.91
mean-field	$\gamma{=}1,\ \delta{=}3$	—



Adoption & its variance over the IUL–MSP plane.